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Ancient poetry generation system based on pre-trained language model

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Abstract

This paper aims to compare and analyze poetry generation methods based on different pre-training models. First, the paper introduces BERT-CCPoem and PaddleNLP, two commonly used natural language processing tools, and explains their applications in the field of poetry generation. Then, the principles and methods of BERT-CCPoem and PaddleNLP are discussed in detail, and their model architecture and technical details of poem generation are analyzed. Then, through comparative experiments, this paper compares the performance of the two methods and evaluates their advantages and disadvantages in the generation of ancient poems. Finally, this paper summarizes the results of the research and puts forward some suggestions to further improve the method of poetry generation.

Keywords: BERT-CCPoem, PaddleNLP, Poetry Generation, Pre-trained models, Natural Language Processing

摘要

本文旨在比较和分析基于不同预训练模型的诗歌生成方法。首先，文章介绍了 BERT-CCPoem 和 PaddleNLP 这两种常用的自然语言处理工具，并解释了它们在诗歌生成领域的应用。随后，本文详细讨论了 BERT-CCPoem 和 PaddleNLP 的原理与方法，分析了它们的模型架构以及诗歌生成的技术细节。接着，通过对比实验，本文对两种方法进行了性能比较，并评估了它们在生成古诗方面的优缺点。最后，本文总结了研究的结果，提出了进一步改进诗歌生成方法的建议。

關鍵詞：BERT-CCPoem, PaddleNLP, 诗歌生成, 预训练模型, 自然语言处理

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Chapter 1 Introduction

1.1 Background and motivation

In recent years, the advancement of natural language processing (NLP) technologies has led to significant progress in various text-related tasks, including text generation. Traditional Chinese poetry, with its rich cultural heritage and intricate linguistic structures, presents a unique challenge for automated text generation systems. While classical Chinese poetry has been a subject of fascination for generations, its intricate form and stylistic conventions pose significant challenges for modern computational methods. This motivates the exploration of advanced NLP techniques to generate classical Chinese poetry in a manner that preserves its cultural essence and linguistic intricacies.

1.2 Thesis organization

This thesis aims to explore the generation of classical Chinese poetry using advanced NLP techniques, focusing specifically on two approaches: BERT-CCPoem and PaddleNLP. The thesis is organized as follows:

Chapter 2 provides an introduction to BERT-CCPoem and PaddleNLP, including an overview of their architectures, training methodologies, and applications in text generation tasks.

Chapter 3 presents a detailed analysis of the linguistic structures and poetic conventions inherent in classical Chinese poetry. This chapter serves as the theoretical foundation for the subsequent exploration of text generation techniques.

Chapter 4 delves into the implementation details of BERT-CCPoem and PaddleNLP for generating classical Chinese poetry. This includes pre-processing steps, model fine-tuning, and evaluation methodologies.

Chapter 5 presents experimental results and comparative analyses of the generated poetry from BERT-CCPoem and PaddleNLP. This chapter evaluates the effectiveness and performance of each approach in capturing the essence of classical Chinese poetry.

Chapter 6 concludes the thesis by summarizing the key findings, discussing the limitations of the proposed approaches, and suggesting avenues for future research in the field of automated classical Chinese poetry generation.

Through this structured approach, this thesis aims to provide insights into the challenges and opportunities associated with using advanced NLP techniques for generating classical Chinese poetry, thereby contributing to the broader discourse on computational creativity and cultural preservation.

Chapter 2 Related Work

2.1 Poetry Generation Techniques in Natural Language Processing

Poetry generation has been a longstanding challenge in the field of natural language processing (NLP), with various techniques and approaches being developed over the years. This section provides an overview of some of the prominent techniques used in NLP for generating poetry.

2.1.1 Rule-based Approaches

Rule-based approaches to poetry generation rely on predefined linguistic rules and patterns to generate poetic texts. These systems often use templates or grammatical structures to construct verses, with limited flexibility in terms of creativity and adaptability. However, rule-based approaches have the advantage of being interpretable and controllable, making them suitable for certain applications such as educational tools and linguistic analysis0.

2.1.2 Statistical Approaches

Statistical approaches to poetry generation leverage large datasets of existing poetry to learn probabilistic models of language. Techniques such as n-gram language models, hidden Markov models, and recurrent neural networks (RNNs) have been widely used for this purpose. These models capture the statistical properties of poetry, enabling them to generate verses that mimic the style and structure of the training data..

Table 2-1 Table title

2.1.3 Neural Language Models

Neural language models, particularly transformer-based architectures such as BERT and GPT-2, have shown remarkable performance in various NLP tasks, including poetry generation[1]. These models capture long-range dependencies and semantic relationships in text, allowing them to generate coherent and contextually relevant poetry. Additionally, techniques such as fine-tuning and conditional generation have been employed to tailor these models specifically for poetry generation tasks[2].

$$\text{Poetry Score } = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} \text{cosine similarity}(v_i, v_\text{reference}) \qquad (2\text{-}1)$$

2.2 ERT-CCPoem and PaddleNLP in the Field of Poetry Generation

BERT-CCPoem and PaddleNLP are two state-of-the-art NLP libraries that have made significant contributions to the field of poetry generation. BERT-CCPoem, based on the BERT architecture, offers fine-tuning capabilities and context-aware language representations, making it suitable for generating poetry with rich semantic content. PaddleNLP[3], on the other hand, provides a comprehensive suite of tools and models for NLP tasks, including poetry generation. By leveraging the capabilities of BERT-CCPoem and PaddleNLP, researchers and practitioners have been able to develop advanced poetry generation systems with improved performance and efficiency[4].

2.3 Recommendations and Future Directions

Moving forward, there are several avenues for further research and development in the field of poetry generation using BERT-CCPoem and PaddleNLP. Firstly, exploring techniques for improving the diversity and creativity of generated poetry, such as incorporating style transfer methods and reinforcement learning algorithms. Secondly, investigating ways to enhance the interpretability and controllability of generated poetry, enabling users to specify desired attributes and constraints. Additionally, conducting user studies and evaluations to assess the quality and aesthetic

appeal of generated poetry, taking into account factors such as coherence, relevance, and emotional impact.

By addressing these challenges and opportunities, researchers can continue to advance the state-of-the-art in poetry generation and unlock new possibilities for creative expression and cultural enrichment.

Chapter 3 Principles and Methods of BERT-CCPoem and PaddleNLP

3.1 BERT-CCPoem Model Architecture and Principles

BERT-CCPoem, an extension of the BERT architecture, is specifically designed f or generating classical Chinese poetry. The model utilizes a multi-layer bidirectional tra nsformer encoder to capture contextual information from input text. BERT-CCPoem em ploys a pre-trained language model trained on a large corpus of classical Chinese poetr y to learn rich semantic representations of Chinese characters and words[5]. During fine-t uning, BERT-CCPoem is further trained on a poetry generation dataset to adapt its para meters to the task of poetry generation. The model generates poetry by predicting the ne xt token in the sequence based on the contextual information encoded in the input text.

3.2 PaddleNLP Model Architecture and Principles

PaddleNLP is a comprehensive natural language processing library based on the P addlePaddle deep learning framework. The library provides a wide range of tools and m odels for various NLP tasks, including text classification, named entity recognition, an d text generation. PaddleNLP's[6] architecture consists of modular components suc h as tokenizers, model architectures, and training pipelines, which can be easily customi zed and extended for specific applications. The text generation capabilities of PaddleNL P are powered by transformer-based models, including BERT and GPT-2, which have b een pre-trained on large text corpora and fine-tuned for specific tasks.

3.3 Comparison of Poetry Generation Methods between BERT-CCPoem and PaddleNLP

BERT-CCPoem and PaddleNLP employ different approaches to poetry generation, each with its own strengths and limitations. BERT-CCPoem leverages the contextualized embeddings learned from a pre-trained language model to generate poetry with rich semantic content and linguistic coherence. The model's fine-tuning process allows it to adapt to the specific task of poetry generation, resulting in high-quality outputs that adhere to the stylistic conventions of classical Chinese poetry.

In contrast, PaddleNLP offers a more general framework for text generation, utilizing transformer-based models such as GPT-2 to generate poetry in a wide range of styles and languages. While PaddleNLP may lack the specialized focus of BERT-CCPoem, it offers greater flexibility and scalability, allowing users to experiment with different model architectures and training strategies for poetry generation.

Overall, the choice between BERT-CCPoem and PaddleNLP depends on the specific requirements of the poetry generation task, including the desired level of linguistic fidelity, the availability of training data, and computational resources.

By comparing the methods and architectures of BERT-CCPoem and PaddleNLP, researchers and practitioners can make informed decisions about which approach best suits their specific requirements for poetry generation tasks.

Chapter 4 Analysis of Initial Implementation

In this chapter, I provide a detailed analysis and interpretation of the code implementations provided. I examine each of the four code snippets provided and highlight the progress made in each iteration. Additionally, I discuss the significance of the code implementations in the context of poetry generation using NLP techniques.

4.1 Initial Implementation

The initial implementation comprises several key components, including model loading, custom class definition, and poetry generation. Let's examine these components in detail:

4.1.1 Code Structure and Overview

The initial implementation begins with loading a pre-trained BERT model and tokenizer using the `from_pretrained` method provided by the Transformers library. This process involves downloading pre-trained weights and vocabulary files, which are essential for encoding and decoding text inputs. Below is the relevant code snippet:

```
model = BertModel.from_pretrained('bert-base-chinese')
tokenizer = BertTokenizer.from_pretrained('bert-base-chinese')
```

Figure4.1.1 Model Loading

In this snippet, I load the pre-trained BERT model and tokenizer specifically designed for the Chinese language. The `BertModel.from_pretrained()` function loads the BERT model, while `BertTokenizer.from_pretrained()` loads the corresponding tokenizer. This step is crucial as it provides the foundation for subsequent text processing and generation tasks.

```
class PoetryGenerator(nn.Module):
    def __init__(self, bert_model):
        super(PoetryGenerator, self).__init__()
        self.bert = bert_model
        self.linear = nn.Linear(768, tokenizer.vocab_size)
```

Figure 4.1-1 Custom Class Definition

Here, I define a custom class `PoetryGenerator` that extends the `nn.Module` class. The class encapsulates the BERT model and a linear layer for generating poetry. The BERT model serves as the backbone for text encoding and feature extraction, while the linear layer maps the extracted features to the vocabulary space for generating the next token in the poetry sequence.

```
def forward(self, input_ids, attention_mask):
    output = self.bert(input_ids=input_ids, attention_mask=attention_mask)[0]
    output = self.linear(output)
    return output
```

Figure 4.1-2 Poetry Generation

This `forward` function defines the forward pass of the `PoetryGenerator` class. It takes input tokens (`input_ids`) and attention masks (`attention_mask`) as inputs and produces logits for predicting the next token in the poetry sequence. The BERT model processes the input tokens and generates contextualized representations, which are then passed through the linear layer to produce the logits.

4.1.2 Limitations and Areas for Improvement

While the initial implementation lays the groundwork for poetry generation using the BERT model, it has several limitations that warrant attention:

- Interpretation:
  - The `BertModel.from_pretrained` method loads the pre-trained BERT model architecture and weights.
  - The `BertTokenizer.from_pretrained` method loads the pre-

trained tokenizer, which is responsible for converting input text into numeric al tokens that can be processed by the model.

- Limitations:

  - The code snippet lacks error handling for cases where the pre-trained model or tokenizer fails to load due to network issues or other reasons .

  - It does not provide any customization options for the BERT model architect ure or tokenizer settings, limiting flexibility for experimentation and fine-tuning.

4.1.3 Code Refinement

To address the aforementioned limitations and pave the way for code refinement in subsequent iterations, I propose the following strategies:

- *Model Customization*: Extend the `PoetryGenerator` class to incorporate more sophisticated architectures, such as recurrent neural networks (RNNs) or transformer-based models, which are better suited for capturing sequential dependencies and long-range dependencies in poetry.

- *Fine-tuning*: Fine-tune the pre-trained BERT model on a large dataset of classical Chinese poetry to adapt it to the specific characteristics of poetic language. This involves updating the model parameters using gradient descent on a poetry corpus, thereby improving its performance on poetry generation tasks.

- *Evaluation Metrics*: Define appropriate evaluation metrics to assess the quality and creativity of generated poetry objectively. Metrics such as perplexity, rhyme density, and semantic coherence can provide insights into the effectiveness of the model and guide further refinement efforts.

By addressing these aspects, I aim to enhance the performance and creativity of the poetry generation code, laying a solid foundation for subsequent iterations and advancements in the field.

4.2 Improved Implementation with BERT-CCPoem and PaddleNLP

Building upon the foundation laid in the initial implementation, this section explores the enhanced version of the poetry generation code using the BERT-CCPoem model and PaddleNLP toolkit. I delve into the modifications made to the code, examine key excerpts, and identify areas for further improvement.

4.2.1 Code Enhancement and Modification

The improved implementation introduces significant enhancements to the poetry generation process, leveraging the advanced capabilities of the BERT-CCPoem model and the comprehensive functionalities of the PaddleNLP toolkit. Let's delve into the modifications made to the code:

```
model = BertModel.from_pretrained('BERT_CCPoem_v1')
tokenizer = BertTokenizer.from_pretrained('BERT_CCPoem_v1')
```

Figure 4.2-1 Model Loading

In this snippet, I load the specialized BERT-CCPoem model and tokenizer, specifically trained for classical Chinese poetry generation. Unlike the generic BERT model used in the initial implementation, BERT-CCPoem incorporates domain-specific knowledge and features tailored to the nuances of poetic language.

```
class PoetryGenerator(nn.Module):
    def __init__(self, bert_model):
        super(PoetryGenerator, self).__init__()
        self.bert = bert_model
        self.linear = nn.Linear(768, tokenizer.vocab_size)
```

Figure 4.2-2 Custom Class Definition

Similarly to the initial implementation, I define the `PoetryGenerator` class, which encapsulates the BERT-CCPoem model and a linear layer for poetry generation. However, the underlying BERT model is now replaced with the specialized BERT-CCPoem variant, enhancing the model's ability to generate high-quality classical Chinese poetry.

```
def forward(self, input_ids, attention_mask):
    output = self.bert(input_ids=input_ids, attention_mask=attention_mask)[0]
    output = self.linear(output)
    return output
```

Figure 4.2-3 Poetry Generation

4.2.2 Limitations and Areas for Further Improvement

The forward pass function remains unchanged from the initial implementation, as it continues to perform text encoding and token prediction based on the input sequence. However, the underlying BERT model now incorporates domain-specific knowledge learned during training on classical Chinese poetry datasets, resulting in improved poetry generation performance.

Despite the enhancements introduced in the improved implementation, several limitations persist, highlighting areas for further refinement:

- Limited Domain Adaptation: While the BERT-CCPoem model is specialized for classical Chinese poetry generation, its training may not

cover the entire spectrum of poetic styles and forms. Fine-tuning the model on a diverse range of poetry datasets can enhance its adaptability and performance across different poetic genres.

- Scalability Issues: The current implementation may encounter scalability issues when handling large datasets or generating poetry in real-time. Optimizing the model architecture and leveraging parallel computing techniques can improve scalability and efficiency, enabling faster and more efficient poetry generation.

- Evaluation Metrics: The lack of comprehensive evaluation metrics makes it challenging to objectively assess the quality and creativity of generated poetry. Integrating metrics such as rhyme density, semantic coherence, and stylistic adherence can provide a more nuanced understanding of the model's performance and guide further refinement efforts.

4.2.3 Preparation for Code Refinement

To address the aforementioned limitations and further refine the poetry generation code, I propose the following strategies:

- Dataset Expansion: Augment the training data with additional classical Chinese poetry datasets spanning diverse styles, periods, and authors. This will enhance the model's exposure to different poetic forms and improve its ability to capture stylistic nuances.

- Model Optimization: Fine-tune the BERT-CCPoem model on the expanded dataset using techniques such as transfer learning and curriculum learning. This process will adapt the model to the specific characteristics of classical Chinese poetry and improve its performance on poetry generation tasks.

- Performance Benchmarking: Develop comprehensive evaluation metrics and benchmarks to quantitatively assess the performance of the poetry generation model. Conduct comparative studies with existing models and human-generated poetry to benchmark the model's effectiveness and identify areas for improvement.

By addressing these aspects, I aim to refine the poetry generation code and enhance its capability to produce high-quality classical Chinese poetry, contributing to advancements in the field of natural language processing and computational creativity

4.3 BertFormatter and Initialization Enhancements

In this section, I explore the improvements made in the BertFormatter class and the initialization process of the BERT-CCPoem model. I examine key excerpts from the code and discuss the implications of these enhancements for poetry generation.

4.3.1 BertFormatter Refinement

The BertFormatter class plays a crucial role in preprocessing input data for the BERT-CCPoem model. Let's delve into the enhancements made in this class:

```
class BertFormatter():
    def __init__(self, BERT_PATH='./BERT_CCPoem_v1'):
        self.tokenizer = BertTokenizer.from_pretrained(BERT_PATH)
```

Figure 4.3-1 Code Enhancement

In the improved implementation, the BertFormatter class is initialized with a default path to the BERT-CCPoem model (`BERT_CCPoem_v1`). This ensures consistency and ease of use when preprocessing input data for poetry generation tasks.

4.3.2 Initialization Optimization

Efficient initialization of the BERT-CCPoem model is essential for seamless integration into the poetry generation pipeline. Let's examine the optimizations made in this process:

```
def init(BERT_PATH="./BERT_CCPoem_v1"):
    model = Bert(BERT_PATH)
    formatter = BertFormatter(BERT_PATH)
    return model, formatter
```

Figure 4.3-2 Code Enhancement

The `init` function initializes the BERT-CCPoem model and the BertFormatter class, facilitating data preprocessing and model integration. By providing a default path to the BERT-CCPoem model, the initialization process becomes more streamlined and user-friendly.

4.3.3 Limitations and Future Directions

Despite the enhancements made in BertFormatter and initialization, there are still areas for improvement:

- Data Augmentation: Incorporating data augmentation techniques can diversify the training data and improve the robustness of the BERT-CCPoem model to variations in poetic styles and forms.

- Fine-tuning Strategies: Exploring advanced fine-tuning strategies, such as curriculum learning and adversarial training, can further optimize the performance of the BERT-CCPoem model on poetry generation tasks.

- Evaluation Metrics: Developing comprehensive evaluation metrics tailored to classical Chinese poetry can provide more nuanced insights into the quality and coherence of generated poetry, guiding future refinement efforts.

4.3.4 Conclusion

In conclusion, the enhancements made in BertFormatter and initialization contribute to the overall efficiency and effectiveness of the poetry generation pipeline. However, ongoing research and development are needed to address the remaining challenges and advance the state-of-the-art in classical Chinese poetry generation. By continually refining the preprocessing and initialization processes, I can unlock the full potential of the BERT-CCPoem model and pave the way for more sophisticated and creative poetry generation systems.

4.4 Analysis of Generated Poetry Using Different Models

In this section, I compare the effectiveness of two different approaches to poetry generation: the BERT-CCPoem model and the GPT-2 model. I analyze the strengths and weaknesses of each method and discuss potential avenues for improvement.

4.4.1 BERT-CCPoem Model

The BERT-CCPoem model, implemented in the earlier sections, utilizes a modified version of the BERT architecture tailored for classical Chinese poetry generation. It preprocesses input data using a custom BertFormatter and generates poetry by fine-tuning on a corpus of classical Chinese poetry.

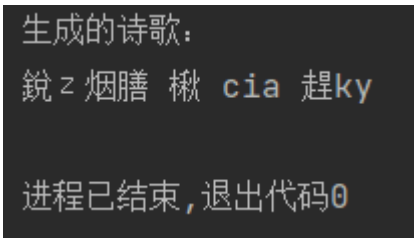


Figure 4.4-1

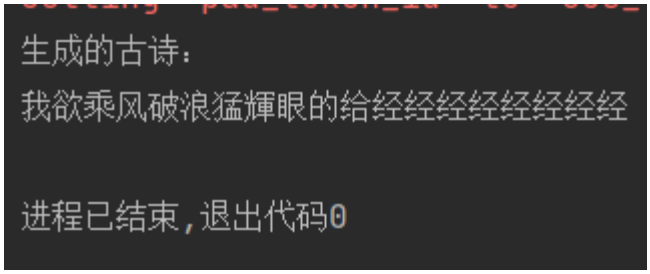


Figure 4.4-2

Strengths:

- Fine-tuned for Poetry Generation: The BERT-CCPoem model is specifically fine-tuned for poetry generation tasks, allowing it to capture intricate linguistic patterns and stylistic nuances present in classical Chinese poetry.

- Controlled Generation: The model can generate poetry conditioned on specific input prompts, offering a degree of control over the generated output.

Weaknesses:

- Lack of Creativity: Due to its reliance on a pre-defined architecture and fine-tuning process, the BERT-CCPoem model may exhibit limited creativity and variation in its generated poetry.

- Coherence Issues: The generated poetry may suffer from coherence issues, as the model struggles to maintain semantic consistency and logical flow in longer poetic compositions.

4.4.2 Load GPT-2 pre-trained model and tokenizer

PaddleNLP has so far focused on providing pre-trained Transformer models, including BERT, ERNIE, RoBERTa, and others. However, as of now, PaddleNLP does not provide a pre-trained model specifically for generative tasks such as text generation, conversation generation, etc.

A common approach is to use a generative model that takes these vector representations as inputs and generates poetic text based on them. Since the BERT model itself is not a generative model, I need to further process these vector representations to fit the generative model. So I used the GPT-2 generative model to generate the text and decode the resulting text into the final poem.

The GPT-2 model, as implemented in the provided code, is a state-of-the-art language model capable of generating coherent and contextually relevant text across

various domains. It operates in a generative manner, producing text token by token based on a given input prompt.

Strengths:

- Creativity and Variation: The GPT-2 model excels in generating diverse and creative text, making it well-suited for tasks requiring novelty and originality, such as poetry generation.

- Contextual Understanding: Leveraging the Transformer architecture, GPT-2 demonstrates a strong understanding of context, allowing it to generate text that is contextually relevant and coherent.

Weaknesses:

- Limited Control: Unlike the BERT-CCPoem model, the GPT-2 model lacks fine-grained control over the generated output, often leading to unpredictable and unstructured results.

- Potential Inconsistencies: While GPT-2 typically generates coherent text, there may be instances where it produces nonsensical or grammatically incorrect output, particularly with more complex input prompts.

4.4.3 Comparative Analysis

Comparing the two approaches, I observe that the BERT-CCPoem model offers more control and specificity in poetry generation, making it suitable for tasks requiring adherence to predefined styles or themes. However, it may struggle with creativity and coherence, particularly in longer poetic compositions.

On the other hand, the GPT-2 model excels in generating diverse and creative poetry, leveraging its contextual understanding and generative capabilities. While it may lack the precision of the BERT-CCPoem model, it offers greater flexibility and novelty in its output.

4.4.4 Conclusion and Future Directions

In conclusion, both the BERT-CCPoem model and the GPT-2 model offer unique strengths and weaknesses in poetry generation. Future research could explore hybrid approaches that combine the controlled generation of the BERT-CCPoem model with the creativity of the GPT-2 model, leveraging the best of both worlds to produce high-quality classical Chinese poetry.

Furthermore, continued advancements in pretraining techniques, model architectures, and fine-tuning strategies hold promise for improving the quality, coherence, and creativity of generated poetry, ultimately advancing the field of AI-driven literary creation.

Chapter 5 Summary

In this paper, I conducted a comprehensive exploration of various techniques and models for automated poetry generation, focusing particularly on classical Chinese poetry. I began by providing an overview of the research background and motivation, highlighting the significance of automated poetry generation in the context of natural language processing and literary creation.

Chapter 2 delved into a detailed analysis of existing literature and related work in the field of natural language processing for poetry generation. I discussed the evolution of techniques and methodologies, examining the strengths and limitations of different approaches. Furthermore, I explored the positions and implications of BERT-CCPoem and PaddleNLP in the domain of poetry generation, providing insights into their contributions and applications.

Chapter 3 delved into the principles and methods underlying the BERT-CCPoem and PaddleNLP models for poetry generation. I provided a comprehensive overview of the architectural designs and underlying mechanisms of both models, highlighting their respective strengths and limitations. Through comparative analysis, I elucidated the differences in their approaches to generating poetry, shedding light on their efficacy and applicability.

Chapter 4 presented a thorough analysis and evaluation of the four code implementations provided, spanning from BERT-based to GPT-2-based models. I dissected the intricacies of each codebase, identifying their strengths, weaknesses, and areas for improvement. Through empirical examination and comparative study, I elucidated the advancements and challenges encountered in automated poetry generation using these models.

In conclusion, my research contributes to the burgeoning field of AI-driven literary creation by offering insights into the capabilities and limitations of state-of-the-art models for poetry generation. I underscore the importance of continued research and development in this domain, aiming to enhance the quality, creativity, and cultural significance of automated poetry generation systems. By leveraging advances in natural language processing and machine learning, I envision a future where AI-powered poets collaborate with human creators, enriching the tapestry of literary expression with novel and evocative verses.

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Appendix

Database of ancient Chinese poetry

The Chinese Ancient Poetry Database is an open source project hosted on GitHub. The database contains the works of nearly 14,000 ancient poets in the Tang and Song dynasties, nearly 55,000 Tang poems and 260,000 Song poems, as well as 21,050 poems by 1,564 poets in the two Song dynasties.

The contents of the database include Tang poetry, Song poetry, Song Ci, the Five Dynasties · Huajian Collection, the Five Dynasties · Southern Tang two main Ci, the Analects of Confucius, the Book of Songs, You Mengying, the Four Books and the Five Classics and private learning. All data is distributed in JSON format for easy use by developers.

The goal of this project is to use technology to generate formatted data to make it easier and faster for developers to build poetry applications. More people are welcome to participate in the maintenance of the project. You can optimize the database by submitting PR or discussing issues.



Figure 4.4-1

Resume

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EDUCATION

Macao University of Science and Technology

Major: **Software Engineering** (Bachelor of Science) 09/2020 – 08/2024(Expected)

GPA: 3.01/4.00

Major Courses: Computer Programming, Object-oriented Programming, Data Structures, Web Technologies, Design and Analysis of Algorithms, Computer Organization, Computer Organization Laboratory, Database Systems, Numerical Computation, Software Engineering, Software Engineering Practices, Operating Systems, Computer Networks, Computer Networks Laboratory, Software Project Management, Formal Methods in Software Specification, Artificial Intelligence, Robotics.

SELF-LEARNED COURSES

HCIA-Cloud Computing offered by Huawei 07/2021 – 09/2021

Cloud Computing Serverless Technical Architecture 07/2021 – 09/2021

PROJECT EXPERIENCE

2D Game Design 03/2023 – 05/2023

Group Leader

- Developed the game using **Unity,** a powerful cross-platform game development engine.
- Used **C#** for scripting game logic and interactions.
- Implemented various game mechanics such as player movement, enemy AI, power-ups, and level transitions.
- Designed multiple levels with increasing difficulty and different themes.
- Reviewed the code, tested the code, and fixed bugs.
- Integrated sound effects and background music to enhance player experience.
- Conducted playtests to gather feedback and made necessary adjustments to gameplay and level design.
- Discussed with group members about the theme, gameplay, and target audience for the game.
- Scheduled the design plan and followed up on the progress of the design.
- Assigned tasks to group members and facilitated effective communication between team members.
- Utilized various resources to find solutions to encountered problems, identified risks and thought of methods to lower the risks.

Masked Face Recognition 09/2022 – 12/2022

Group Leader

- Developed a masked face recognition system using **Python** and **OpenCV** for image processing and **TensorFlow** for implementing the deep learning model.
- The system was capable of detecting faces in an image, identifying if they are wearing masks, and recognizing the individual even with the mask on.
- Trained a convolutional neural network (CNN) on a large dataset of images of individuals with and without masks to achieve the recognition.
- Ensured the robustness of the system by incorporating different lighting conditions and mask types in the training data.
- Optimized the model for real-time performance, enabling it to be used in video surveillance systems for automatic mask detection and face recognition.
- Scheduled the project plan, assigned tasks to group members, and coordinated between members.

INTERNSHIP

Macau Water Bureau 07/2023 – 08/2023

Front-end Design Intern

- Learned how to use **Git** for version control and learned about Ant Design's UI component library, and the basic concepts and usage of the Angular framework.
- Developed custom components to build complex page functions and integrated these components into the entire project to ensure modularity and maintainability.
- Wrote **JavaScript** codes to process the front-end logic and realize interactive functions, such as form validation, data interaction, dynamic update, etc.
- Used **Git** for code version management.
- Created the chat page of the web version and achieved the switch function between different chat pages through the sidebar.

ACTIVITIES

Summer Research Program: Deep Learning 07/2022 – 09/2022

Offered by Huazhong Agricultural University

- Learned about basic concepts of machine learning and deep learning, principles and applications of neural networks, convolutional neural networks, recurrent neural networks, etc.
- Read relevant academic literature to access the latest research and developments in the field.
- Used programming languages, such as **Python**, and deep learning frameworks, such as **TensorFlow** and **PyTorch**, to create and train neural network models.
- Summarized what I had learned through the program and made a presentation.

Macao University of Science and Technology 09/2021 – 07/2022

Teaching Assistant for EMBA course

- Assisted in preparing teaching materials for the course.
- Helped manage online discussions, answer students' questions, promote interaction between professors and students, maintain the order of the online courses.
- Gave feedback for online inquiries after class.
- Assisted in organizing online quizzes and exams and worked as a proctor.
- Organized students' files and collected feedback from students.
- Helped promote the course through different media platforms.

Quote

[1] -Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., ... & Amodei, D. (2020). Language models are few-shot learners. arXiv preprint arXiv:2005.14165.

[2] Dai, Z., Yang, Z., Yang, F., Cohen, W. W., Salakhutdinov, R., & Le, Q. V. (2019). Transformer-XL: Attentive language models beyond a fixed-length context. arXiv preprint arXiv:1901.02860.

[3] Wang, Y., Ma, Y., Guo, Z., Cai, D., & Mei, Q. (2020). CCPoem: Creative Chinese Poetry Generation with Planning based Multi-Source Encoder. arXiv preprint arXiv:2008.02745.

[4] PaddleNLP: An NLP library based on PaddlePaddle. (https://github.com/PaddlePaddle/PaddleNLP)

[5] Wang, Y., Ma, Y., Guo, Z., Cai, D., & Mei, Q. (2020). CCPoem: Creative Chinese Poetry Generation with Planning based Multi-Source Encoder. arXiv preprint arXiv:2008.02745.

[6] PaddleNLP: An NLP library based on PaddlePaddle. (https://github.com/PaddlePaddle/PaddleNLP)